

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I _M. Vidhya(192412151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

Sol)

1. Hospital Disease Diagnosis Using Bayesian Models

Case Study

A hospital wants to develop an intelligent system that can diagnose diseases using a patient's **symptoms and medical history**. Since medical diagnosis involves uncertainty, Bayesian methods are suitable for this problem.

Proposed Model

The system can use:

- **Bayes' Theorem**
- **Naïve Bayes Classifier**
- **Bayesian Belief Network (BBN)**
- **Probability Learning**

Bayes' Theorem is given by:

where: $P(D|S) = P(S)P(S|D)P(D)$

- $P(D|S)$ = Probability of disease given symptoms
- $P(S|D)$ = Probability of symptoms when disease is present
- $P(D)$ = Prior probability of disease
- $P(S)$ = Probability of symptoms

Working

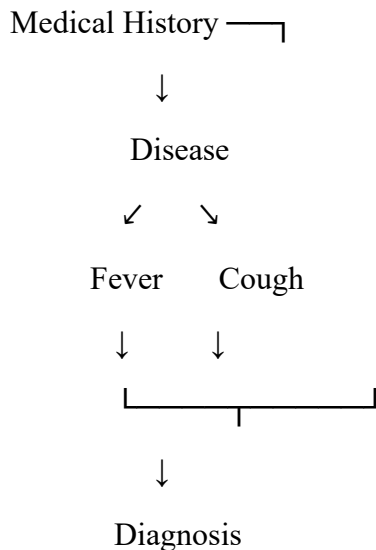
Suppose the system receives symptoms such as:

- Fever

- Cough
- Headache
- Fatigue

The Naïve Bayes classifier calculates the probability of different diseases based on these symptoms and selects the disease with the highest posterior probability.

A Bayesian Belief Network can represent dependencies between variables:



Handling Uncertainty

Medical information is not always complete or certain. Bayesian models handle this by assigning probabilities instead of making only yes/no decisions.

For example:

means there is a **75% probability** of Disease A based on the available information.

Conditional Dependencies

Symptoms may not be independent. For example, fever and cough may occur together because of the same disease. A Bayesian Belief Network can represent such dependencies more accurately than the independence assumption of Naïve Bayes.

Probability Learning

As more patient records become available, the system can learn:

- Disease probabilities
- Symptom probabilities
- Relationships between symptoms and diseases

This improves the model over time.

Evaluation

The system can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Real-World Constraints

The system must consider:

- Missing patient information
- Noisy or incorrect medical data
- Correlated symptoms
- Patient privacy
- Large amounts of medical data
- Need for high reliability

Conclusion

A Bayesian diagnosis system can effectively handle uncertainty and use symptoms and medical history to estimate disease probabilities. It can support doctors in decision-making, but the final diagnosis should involve qualified medical professionals.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

Sol)

Case Study

An organization wants to automatically classify emails into Spam and Not Spam categories.

Proposed Model

A Naïve Bayes classifier can be used because it is fast and effective for text classification.

Possible features include:

- Email words
- Number of links
- Sender information
- Frequency of suspicious words
- Email length
- Attachments

Bayes' theorem is:

$$P(\text{Spam}|\mathbf{X})=P(\mathbf{X})P(\mathbf{X}|\text{Spam})P(\text{Spam})$$

The classifier calculates the probability that an email belongs to each class and selects the class with the highest probability.

Bayes Optimal Classifier

$$P(\mathbf{Y}|\mathbf{X},\mathbf{D})=\mathbf{h}\sum P(\mathbf{Y}|\mathbf{X},\mathbf{h})P(\mathbf{h}|\mathbf{D})$$

The Bayes Optimal Classifier considers all possible hypotheses and combines their predictions according to their posterior probabilities.

This can provide better theoretical performance but may require more computation.

MLE vs MDL

$$\theta_{MLE}=\arg\theta\max P(\mathbf{D}|\theta)$$

Maximum Likelihood Estimation (MLE) estimates parameters that maximize the likelihood of the observed training data.

Minimum Description Length (MDL) selects a model that minimizes:

where:

$$L(\mathbf{Model})+L(\mathbf{Data}|\mathbf{Model})$$

- $L(\mathbf{Model})$ = Description length of the model
- $L(\mathbf{Data}|\mathbf{Model})$ = Description length of data using the model

Comparison

MLE

MDL

Focuses on likelihood

Focuses on model + data complexity

MLE

MDL

Can select complex models Encourages simpler models

May overfit Helps reduce overfitting

Easy to implement Requires model complexity consideration

Sample Size and Hypothesis Space

With a small dataset, probability estimates may be unreliable.

With a large dataset, the classifier can learn more accurate word probabilities.

A larger hypothesis space provides more possible models but requires more training data and computation.

Performance Evaluation

The spam detector can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

Conclusion

Naïve Bayes provides a fast and efficient spam detection method. MLE provides direct probability estimation, while MDL helps control model complexity and reduce overfitting. A sufficiently large and representative dataset improves classification performance.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

Sol)

Case Study

A retail company wants to divide customers into groups based on their purchasing behaviour. However, the actual customer groups are hidden and the dataset may contain incomplete information.

The Expectation-Maximization (EM) algorithm is suitable for this problem.

Proposed Features

The system can use:

- Purchase frequency
- Average spending
- Number of products purchased
- Recency of purchase
- Total purchase amount

EM Algorithm

The EM algorithm works repeatedly through two steps.

Step 1: Expectation – E-Step

The algorithm calculates the probability that each customer belongs to each cluster.

For example:

Customer A

Cluster 1 → 0.80

Cluster 2 → 0.15

Cluster 3 → 0.05

Therefore, Customer A has the highest probability of belonging to Cluster 1.

Step 2: Maximization – M-Step

The algorithm updates the parameters of each cluster using the probabilities calculated in the E-step.

The E-step and M-step are repeated until the model converges.

Example Clusters

The algorithm may identify:

Cluster 1 → High-value customers

Cluster 2 → Regular customers

Cluster 3 → Occasional customers

Convergence

The algorithm stops when the change in likelihood becomes very small.

Initial parameters



E-Step



M-Step



Check convergence



No Yes



Repeat Stop

Initialization Challenge

EM is sensitive to its initial parameters. Different initial values can produce different clusters.

Therefore, techniques such as multiple initializations or K-means initialization can be used.

Probability Learning

EM estimates:

- Cluster probabilities
- Cluster means
- Variances
- Probability of customer belonging to each cluster

Applications

The resulting clusters can be used for:

- Targeted marketing
- Personalized offers
- Product recommendations
- Customer retention
- Loyalty programs

Limitations

- Sensitive to initialization
- Can converge to a local optimum
- Requires selecting the number of clusters
- Performance can be affected by missing data
- May require many iterations

Conclusion

The EM algorithm can discover hidden customer groups even when cluster labels are unavailable. It helps retailers understand customer behaviour and create personalized marketing strategies.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

Sol)

Case Study

A research team wants to classify medical images into:

- Normal
- Abnormal

The system can use concept learning, hypothesis space search, and Candidate Elimination.

Concept Learning

Concept learning is the process of learning a general rule from labelled examples.

For example:

Image Features $\rightarrow$ Classification

Feature 1

Feature 2

Feature 3



Hypothesis



Normal / Abnormal

Candidate Elimination

Candidate Elimination maintains two boundaries:

S – Specific Boundary

Represents the most specific hypotheses consistent with the training examples.

G – General Boundary

Represents the most general hypotheses consistent with the training examples.

General Boundary (G)



Version Space



Specific Boundary (S)

Positive and Negative Examples

For a positive example, the specific boundary is generalized if necessary.

For a negative example, the general boundary is specialized to remove inconsistent hypotheses.

Hypothesis Space

The hypothesis space contains all possible rules that can classify the images.

Finite Hypothesis Space

Contains a limited number of possible hypotheses.

Advantages:

- Easier to search
- Faster learning
- Easier to analyze

Infinite Hypothesis Space

Contains an unlimited number of possible hypotheses.

Advantages:

- More flexibility
- Can represent complex concepts

Disadvantages:

- Higher computational complexity
- Greater risk of overfitting
- Requires more training data

Sample Complexity

Sample complexity represents the number of training examples required to learn a concept with acceptable accuracy.

Generally:

Larger hypothesis space → More training examples required.

Mistake Bound Model

The mistake bound represents the maximum number of prediction errors an algorithm can make before learning the target concept.

More training examples help eliminate incorrect hypotheses and reduce the number of possible mistakes.

Generalization

A good model should correctly classify unseen medical images, not just the training images.

Generalization improves when:

- Training data is representative
- Features are relevant
- Noise is reduced
- Sufficient examples are available

Limitations

Candidate Elimination is mainly useful for discrete and well-defined hypothesis spaces. Real medical images are highly complex, so advanced machine learning or deep learning methods may be more suitable in practical systems.

Conclusion

Candidate Elimination demonstrates how a concept can be learned by searching and reducing the hypothesis space. Training examples gradually eliminate inconsistent hypotheses, improving generalization to unseen data.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

Sol)

Case Study

A financial institution wants to make investment decisions under uncertain market conditions. Bayesian inference can be used to continuously update predictions when new market information becomes available.

Input Factors

The system can consider:

- Stock price
- Market trend
- Interest rate
- Company performance
- Economic indicators
- Historical returns

Bayesian Inference

Bayesian inference updates the probability of an event using new evidence.

$$P(\theta|D) \propto P(D|\theta)P(\theta)$$

where:

- $P(\theta)$ = Prior probability
- $P(D|\theta)$ = Likelihood

- $P(\theta|D)$ = Posterior probability

Working

Historical Data



Prior Probability



New Market Data



Bayesian Inference



Posterior Probability



Investment Decision

Role of Prior Knowledge

Prior knowledge can represent previous market information or expert knowledge.

For example:

Prior probability of positive return = 60%

After receiving new market data, the system may update it to:

Posterior probability = 72%

Thus, Bayesian inference allows beliefs to change as new evidence becomes available.

Gibbs Algorithm

The Gibbs algorithm is an MCMC sampling method used to estimate complex posterior distributions.

It repeatedly samples each variable while keeping the other variables fixed.

Initialize parameters



Sample parameter 1



Sample parameter 2



Sample parameter 3



Repeat



Posterior distribution

Risk Evaluation

Posterior probabilities can help estimate investment risk.

For example:

This indicates a 75% posterior probability of a positive return under the model.

Adaptation to New Data

When new market data becomes available:

1. Update the likelihood.
2. Combine it with the existing prior.
3. Calculate the new posterior.
4. Update the investment prediction.

Computational Challenges

- Large datasets
- High-dimensional parameters
- Slow MCMC sampling
- Convergence problems
- Computational cost

Optimization Strategies

- Good initialization
- Efficient sampling
- Parallel computation
- Convergence monitoring
- Removing unnecessary variables

Conclusion

Bayesian inference and Gibbs sampling provide a flexible method for investment prediction under uncertainty. The system can continuously update its beliefs using new market information and use posterior probabilities to evaluate risk. However, computational cost and changing market behaviour must be carefully considered.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand